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Inventor(s)

Von Oepen; Randolph et al.

Percutaneous Delivery of Catheter

Abstract

A delivery system for delivering a prosthetic heart valve, the delivery system including: a handle assembly; a catheter assembly extending from the handle assembly, the catheter assembly having a first catheter and a second catheter positioned radially inside the first catheter; a connecting ring fixed to a distal end portion of the first catheter; and a valve retainer fixed to a distal end portion of the second catheter, the valve retainer configured to receive at least a portion of the prosthetic heart valve therein, wherein the catheter assembly has i) a locked state in which the valve retainer is axially locked relative to the connecting ring, and ii) an unlocked state in which the valve retainer is axially movable relative to the connecting ring, the handle assembly being configured to actuate the catheter assembly between the locked state and the unlocked state.

Inventors: Von Oepen; Randolph (Aptos, CA), Hake; Jarred (Fremont, CA), Sherman; Austin (Sunnyvale, CA)

Applicant: Cephea Valve Technologies, Inc. (Abbott Park, IL)

Family ID: 1000008347960

Assignee: Cephea Valve Technologies, Inc. (Abbott Park, IL)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Ser. No. 63/561,838, filed Mar. 6, 2024, the disclosure of which is hereby incorporated by reference in its entirety as if fully set forth herein.

BACKGROUND

[0002] The present disclosure relates generally to devices, systems and methods for delivering an interventional device into a patient for implantation.

[0003] Intravascular medical procedures allow the performance of therapeutic treatments in a variety of locations within a patient's body while requiring only relatively small access incisions. An intravascular procedure may, for example, eliminate the need for open-heart surgery, reducing risks, costs, and time associated with an open-heart procedure. The intravascular procedure also enables faster recovery times with lower associated costs and risks of complications. An example of an intravascular procedure that significantly reduces procedure and recovery time and cost over conventional open surgery is a heart valve replacement or repair procedure in which an artificial valve or valve repair device is guided to the heart through the patient's vasculature. For example, a catheter is inserted into the patient's vasculature and directed to the inferior vena cava. The catheter is then urged through the inferior vena cava toward the heart by applying force longitudinally (e.g., pushed forward) to the catheter. Upon entering the heart from the inferior vena cava, the catheter enters the right atrium. The catheter may be guided across the atrial septum (e.g., via a guidewire that has already been passed through the atrial septum) into the left atrium. The distal end of the catheter may be deflected by one or more deflecting mechanisms in order to align the distal end of the catheter, as well as a medical device positioned therein, with the mitral valve. Catheter deflection can be achieved by tension cables, or other mechanisms positioned inside the catheter. Precise control of the distal end of the catheter allows for more reliable and faster positioning of a medical device and/or implant and other improvements in the procedures.

BRIEF SUMMARY

[0004] One aspect of the present disclosure provides a delivery system for delivering a prosthetic heart valve, the delivery system comprising: a handle assembly; a catheter assembly extending from the handle assembly, the catheter assembly having a first catheter and a second catheter positioned radially inside the first catheter; a connecting ring fixed to a distal end portion of the first catheter; and a valve retainer fixed to a distal end portion of the second catheter, the valve retainer configured to receive at least a portion of the prosthetic heart valve therein, wherein the catheter assembly has i) a locked state in which the valve retainer is axially locked relative to the connecting ring, and ii) an unlocked state in which the valve retainer is axially movable relative to the connecting ring, the handle assembly being configured to actuate the catheter assembly between the locked state and the unlocked state.

[0005] In one example, the first catheter has a coiled or braided section positioned proximal to the connecting ring.

[0006] In one example, the second catheter has a greater resistance to longitudinal compression compared to the first catheter.

[0007] In one example, the first catheter is an outer sheath of the catheter assembly; and the second catheter is a steering catheter or an extension catheter.

[0008] In one example, the first catheter is an outer sheath of the catheter assembly; and the second

catheter a steering catheter or an intermediate catheter.

[0009] In one example, the connecting ring defines a slot configured to receive a tab of the valve retainer.

[0010] In one example, the valve retainer defines a slot configured to receive a tab of the connecting ring.

[0011] In one example, the handle assembly includes a locking assembly such that rotation of the locking assembly relative to other components of the handle assembly actuates the catheter assembly between the locked state and the unlocked state.

[0012] In one example, the catheter assembly includes a valve cover fixed to the connecting ring, the valve cover configured to maintain at least part of the prosthetic heart valve in a collapsed condition.

[0013] In one example, in the locked state of the catheter assembly, the valve cover is axially locked relative to the valve retainer, thereby preventing premature expression of the prosthetic heart valve from the valve cover.

[0014] In one example, in the unlocked state of the catheter assembly, the valve cover is configured to retract proximally along with the first catheter relative to the second catheter and relative to the valve retainer, thereby allowing the for expression of the prosthetic heart valve from the valve cover.

[0015] In one example, the valve cover is threadedly coupled to the connecting ring.

[0016] In one example, the wherein the handle assembly includes (i) a valve cover retraction knob and (ii) a locking assembly, such that rotation of the valve cover retraction knob in the unlocked state of the catheter assembly proximally withdraws the valve cover relative to the valve retainer, and rotation of the locking assembly relative to other components of the handle assembly actuates the catheter assembly between the locked state and the unlocked state, the valve cover retraction knob being positioned distal to the locking assembly.

[0017] In one example, the valve retainer is threadedly coupled to the second catheter.

[0018] Another aspect of the disclosure provides a method of delivering a prosthetic heart valve, comprising: advancing a catheter assembly through a vasculature while the catheter assembly is in a locked state in which a first catheter is axially locked relative to a second catheter nested within the first catheter, and while the prosthetic heart valve is maintained in a collapsed condition within a valve cover coupled to a distal end of the first catheter; actuating the catheter assembly to an unlocked state, thereby allowing for axial movement of the first catheter relative to the second catheter; and retracting the valve cover and the first catheter relative to the second catheter to deploy the prosthetic heart valve into a native mitral valve annulus.

[0019] In one example, actuating the catheter assembly to the unlocked state is performed when a distal end of the catheter assembly is positioned within or adjacent to an opening of an inferior vena cava leading to a right atrium.

[0020] In one example, actuating the catheter assembly to the unlocked state includes rotating a locking assembly of a handle assembly, the catheter assembly extending from the handle assembly.

[0021] In one example, the catheter assembly includes a connecting ring that couples the valve cover to the distal end of the first catheter, and a valve retainer coupled to a distal end of the second catheter.

[0022] In one example, actuating the catheter assembly to the unlocked state includes rotating the first catheter relative to the second catheter such that the valve retainer rotates relative to the connecting ring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] A more complete appreciation of the subject matter of the present disclosure and the various advantages thereof can be realized by reference to the following detailed description in which reference is made to the accompanying drawings in which:

[0024] FIG. 1 is a perspective view of a delivery system configured for delivering, positioning and deploying a prosthetic heart valve, including a handle assembly and a catheter assembly;

[0025] FIG. 2 is an exploded view of a portion of the handle assembly of the delivery system of FIG. 1, including valve cover retraction knob and cone assembly;

[0026] FIG. 3 is a side view of a portion of the catheter assembly of the delivery system of FIG. 1, including a catheter shaft, coiled and/or braided section, valve cover, and nosecone;

[0027] FIG. 4A is a transverse cross-sectional view of the catheter shaft of FIG. 3, showing various components thereof;

[0028] FIGS. 4B-4C are transverse cross-sectional views of a catheter assembly, operable with the delivery system of FIG. 1, according to another aspect of the disclosure;

[0029] FIG. 5A is a longitudinal cross-section of the catheter assembly of FIG. 3, showing the nested arrangement of various components thereof;

[0030] FIG. 5B is a longitudinal cross-section of a catheter assembly, operable with the delivery system of FIG. 1 and according to another aspect of the disclosure, showing the nested arrangement of various components thereof;

[0031] FIG. 6 is a side view of two catheter assemblies, one catheter assembly in an uncompressed condition and another catheter assembly in a longitudinally compressed condition;

[0032] FIG. 7A is a perspective view of a connecting ring according to one or more aspects of the disclosure;

[0033] FIG. 7B is a perspective view of a valve retainer according to one or more aspects of the disclosure;

[0034] FIG. 7C is a longitudinal cross-section of the connecting ring of FIG. 7A coupled with the valve retainer of FIG. 7B;

[0035] FIG. 8A is a perspective view of a connecting ring according to one or more aspects of the disclosure;

[0036] FIG. 8B is a perspective view of a valve retainer according to one or more aspects of the disclosure;

[0037] FIGS. 8C-D are transverse cross-sectional views of the connecting ring of FIG. 8A coupled with the valve retainer of FIG. 8B;

[0038] FIG. 9A is a perspective view of a connecting ring according to one or more aspects of the disclosure;

[0039] FIG. 9B is a front view of the connecting ring of FIG. 9A according to one or more aspects of the disclosure;

[0040] FIG. 9C is a rear view of the connecting ring of FIG. 9A according to one or more aspects of the disclosure;

[0041] FIG. 9D is a perspective view of a valve retainer according to one or more aspects of the disclosure;

[0042] FIG. 9E is a front view of the valve retainer of FIG. 9D according to one or more aspects of the disclosure;

[0043] FIG. 9F is a rear view of the valve retainer of FIG. 9D according to one or more aspects of the disclosure;

[0044] FIG. 9G is a perspective view of the connecting ring of FIGS. 9A-C coupled with the valve retainer of FIGS. 9D-F according to one or more aspects of the disclosure;

[0045] FIG. 9H is a transverse cross-sectional view of the connecting ring of FIGS. 9A-C coupled with the valve retainer of FIGS. 9D-F according to one or more aspects of the disclosure;

[0046] FIG. 9I is a longitudinal cross-sectional view of the connecting ring of FIGS. 9A-C coupled with the valve retainer of FIGS. 9D-F according to one or more aspects of the disclosure; [0047] FIG. 10A is a perspective view of a cone assembly operable with the delivery system of FIG. 1; [0048] FIGS. 10B-C are side views of the handle assembly of FIG. 10A in unlocked and locked states, respectively; [0049] FIG. 11 is a schematic cutaway view of the heart, showing an exemplary approach for delivering a prosthetic heart valve to a mitral valve annulus; and [0050] FIG. 12 is a flow chart depicting example steps of a method of operating a handle assembly during delivery of the prosthetic heart valve to the mitral valve annulus.

DETAILED DESCRIPTION

[0051] As used herein, the term “inflow end,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve through which blood enters when the heart valve is functioning as intended, whereas the term “outflow end,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve through which blood exits when the heart valve is functioning as intended. For a prosthetic mitral valve, the inflow end is closest to the left atrium when the heart valve is implanted in a patient, and the outflow end is closest to the left ventricle when the heart valve is implanted in a patient. Further, when used herein in connection with a delivery device, the terms “proximal” and “distal” are to be taken as relative to a user operating the device in an intended manner. “Proximal” is to be understood as relatively close to the user and “distal” is to be understood as relatively farther away from the user. Also as used herein, the terms “substantially,” “generally,” and “about” are intended to mean that slight deviations from absolute are included within the scope of the term so modified.

[0052] Native mitral and tricuspid valves, especially the native tricuspid valve, are typically larger in diameter than a native aortic valve. Thus, collapsible and expandable prosthetic mitral and tricuspid valves are typically much larger in diameter than collapsible and expandable prosthetic aortic and pulmonary valves. As a result, transcatheter delivery devices for prosthetic mitral and tricuspid valves are typically larger in size than transcatheter delivery devices for prosthetic aortic and pulmonary valves. It is generally desirable for transcatheter delivery devices to be as small as possible while still being capable of retaining the prosthetic heart valve therein and otherwise functioning as intended. Some transcatheter prosthetic heart valve delivery devices are used with separate introducers that are first inserted into the relevant vasculature (e.g., the femoral vein), and only after the introducers are in place, the delivery device is inserted into the vasculature through the introducer. In some instances, the introducers are necessary or at least important because they may prevent bleeding during the delivery procedure since the introducer typically has a hemostatic valve at its proximal end and provides a sealing function when no device is inserted into the delivery device. The hemostatic valve can also provide a sealing function around devices inserted into the vasculature during the procedure, such as guide wires, catheters, device(s) to cross the native septum, etc. The introducer is stagnant during the procedure, relative to the vasculature, such that the vasculature may be shielded against motion needed during the procedure (e.g., advancing, retraction, and/or rotation of a catheter). The introducer can allow for a catheter exchange. Further, the introducer may allow the delivery device to avoid direct contact with the patient's vasculature for at least some length of the delivery route. As described in greater detail below, one potential benefit of using an introducer to avoid direct contact between the delivery device and the patient's vasculature is that friction between the components may cause negative effects on the delivery device itself, for example causing longitudinal compression that might result in premature deployment of a prosthetic heart valve maintained therein. However, one negative effect of using an introducer is that, because the delivery device must pass through the introducer, the total size of the delivery device and introducer is larger than if the delivery device were able to be used without the introducer. Also, in some examples, a cut-down of the blood vessel will need to be performed to

accommodate the overall larger diameter of the introducer. Thus, it may be desirable to be able to eliminate the need for a separate introducer and directly insert the delivery device into the vasculature. This may result in either (i) a reduction of the total size of the system passing into the patient's vasculature or (ii) the ability to increase the size of the delivery system (which may be beneficial) while maintaining the same overall diameter of the system because there is a concomitant reduction in size in view of the omission of the introducer. However, in order to eliminate the use of the introducer, the delivery device may need to be modified so that the typical benefit(s) of using the introducer are no longer needed. The disclosure provided herein may, in some instances, address one or more of these issues.

[0053] FIG. 1 is a perspective view of a delivery system **100** configured for delivering, positioning and deploying a prosthetic heart valve, including a handle assembly **105** and a catheter assembly **150**. FIG. 2 is an exploded view of a portion of the handle assembly **105** of the delivery system **100** of FIG. 1, including valve cover retraction knob **115** and cone assembly **120** (which may also be referred to as a locking assembly herein). FIG. 3 is a side view of a distal end portion of the catheter assembly **150** of the delivery system **100** of FIG. 1, including a catheter shaft **155**, coiled and/or braided section **160**, valve cover **165** (which may also be referred to as a valve capsule), and an atraumatic distal tip or nosecone **170**.

[0054] Delivery system **100** generally includes a handle assembly **105** and a catheter assembly **150**. Catheter assembly **150** extends from a proximal end coupled to handle assembly **105** to an atraumatic tip or nosecone (not shown in FIG. 1) at a distal end and includes a plurality of catheter and/or hypotube components, at least some of which are longitudinally slidable relative to one another and provide different functionality during operation of delivery system **100** to enable effective delivery and deployment of a prosthetic heart valve, such as a prosthetic mitral valve. While the catheter assembly **150** is depicted as having a length relative to handle assembly **105**, it should be understood that the catheter assembly **150** can have any suitable length and, in some examples, has a much greater length than the handle assembly **105**.

[0055] Attached to a housing **105a** of the handle assembly **105** can be various control mechanisms for controlling one or more aspects of the delivery. For example, attached to the housing **105a** of the handle assembly **105** can be one or more knobs **110** for controlling steering of the of the catheter assembly **150**. Additional details regarding steering operations that may be utilized in conjunction with the components and features described herein are described in U.S. Patent Application Publication No. 2023/0364387, filed Apr. 19, 2023, entitled "Advanced 3-Way Steering," the disclosure of which is hereby incorporated by reference herein. The handle assembly **105** can also include a valve cover retraction knob **115** and a locking or cone assembly **120**. The valve cover retraction knob **115** can be rotated to allow for retraction of valve cover **165** during delivery and/or expression of a prosthetic heart valve. In some examples, rotating valve cover retraction knob **115** results in proximal translation of a catheter member that is attached to the valve cover **165**, allowing for the valve cover **165** to uncover a self-expandable prosthetic heart valve to allow the prosthetic heart valve to self-expand. The cone assembly **120** can be rotated relative to housing **105a**, as is explained in greater detail below, to lock or unlock an outer catheter relative to an inner catheter.

[0056] The catheter assembly **150** can include a catheter shaft **155**, a coiled and/or braided section **160**, a valve cover **165**, and a nosecone (not shown in FIG. 1), as will be described in greater detail below. It should also be understood that catheter assembly **150** may include a number of additional catheter members nested within catheter shaft **155**, as described in greater detail below.

[0057] FIG. 4A is a transverse cross-sectional view of the catheter shaft **155** of FIG. 3, showing various components thereof, including addition catheter components nested therein.

[0058] As shown in FIG. 4A, which is a transverse cross-sectional view of catheter assembly **150** taken along line 4-4 of FIG. 3, these components may include an outer sheath **405a**, a steering catheter **410a**, an extension catheter **415a**, a suture catheter **420a**, and a nosecone catheter **425a**, all

arranged in a concentric nested relationship. The arrangement of these components, as well as valve cover **550a**, nosecone **575a**, and valve retainer **590a** (also referred to as “can”) is shown in the longitudinal cross-section of catheter assembly **150** shown in FIG. 5A. During delivery, the proximal petals of the compressed prosthetic valve can be nested inside of the valve retainer when the prosthetic valve is held back by the suture catheter, which is under tension. In one example, valve retainer **590a** may be attached directly to a tip ring of the steering catheter **410a**. As illustrated in the figures, nosecone catheter **425a** has a lumen sized to receive a guidewire **430a** therein. Each of these components is described in detail below. It should be understood that the valve cover **165** of FIG. 1 and FIG. 3 may be substantially the same as or identical to the other valve covers described herein. Similarly, it should be understood that the atraumatic distal tip or nosecone **170** of FIG. 3 may be substantially the same as or identical to the other atraumatic distal tips or nosecones described herein. In some examples, the suture catheter **420a** may have a plurality of sutures or other wire loops attached to a distal end thereof, the suture loops or wire loops configured to selectively maintain a connection between the prosthetic heart valve and the delivery device.

[0059] To selectively control the curvature of a distal section of steering catheter **410a**, the steering catheter **410a** may be provided with a plurality of tension cables (not shown). The tension cables may travel from handle assembly **105** (e.g., one or more of the knobs **110**) through a plurality of lumens **412a** (which in some embodiments may be polymer tubes, such as Nylon, Pebax, Polyimide, or polytetrafluoroethylene or “PTFE” tubes, or any other type of polymer) to the distal end of steering catheter **410a**. In one embodiment, steering catheter **410a** may include four lumens **412a** equally spaced at 90° intervals around the circumference of steering catheter **410a**. In the embodiment of FIG. 4A, steering catheter **410a** may include four pairs of lumens **412a** (a total of eight lumens **412a**), with the pairs of lumens equally spaced at 90° intervals around the circumference of steering catheter **410a**. Any number of these lumens **412a** (or pairs of lumens) may be provided depending on the various directions of deflection that may be desired. For example, six pairs of lumens **412a** may be provided, with each pair of lumens **412a** housing one steering cable of a pair of steering cables coupled to a corresponding steering knob **110** (e.g., two steering cables attached to each steering knob **110**, with each steering cable extending through one pair of lumens **412a**, for a total of twelve lumens **412a** arranged in six pairs).

[0060] FIGS. 4B-4C are transverse cross-sectional views of a catheter assembly **150b**, operable with the delivery system of FIG. 1 (e.g., in place of catheter assembly **150**), according to another aspect of the disclosure.

[0061] In this example, catheter assembly **150b** omits an extension catheter (e.g., extension catheter **415a** described above) and includes an outer catheter **405b**, a steering catheter **410b**, an intermediate catheter **415b**, a suture catheter **420b**, and a nosecone catheter **425b**. Also in this example, the steering catheter **410b** includes six pairs (a total of 12) of lumens **412b**, which again may be formed with polymer tubes. In this example, the extension catheter is omitted and an intermediate catheter **415b** is implemented in its place. Implementation of the intermediate catheter **415b** can advantageously reduce an overall diameter (e.g., French size) of the catheter assembly **150**. A distal portion of the intermediate catheter **415b** can be fixed (e.g., by a threaded connection) to a distal tip of the steering catheter **410b**. The distal portion of the intermediate catheter **415b** can have a portion of greater diameter, while the proximal portion can be adjacent to the suture catheter **420b** to allow for smooth advancement/retraction of the suture catheter **420b** during deployment.

[0062] The arrangement of these components, as well as valve cover **550b**, nosecone **575b**, and valve retainer **590b** is shown in the longitudinal cross-section of catheter assembly **150b** shown in FIG. 5B.

[0063] FIG. 6 is a side view of two catheter assemblies **600a**, **600b**, one catheter assembly **600a** in an uncompressed condition and another catheter assembly **600b** in a longitudinally compressed condition. As shown in FIG. 6, the coil/braid **605a**, **605b** (which may be substantially similar to

coiled and/or braided section **160**) may have very little resistance against compression. When the force to insert the catheter is larger than the release force of the valve, the coil/braid **605a**, **605b** may be longitudinally compressed and the prosthetic heart valve maintained in the valve cover (e.g., valve cover **165**) may be prematurely released during delivery. As noted above, this potential problem may be largely or entirely avoided by the use of a separate introducer which is first inserted into the patients' vasculature, which may result in the frictional forces between the delivery device and the introducer during insertion to be lower than the release forces of the prosthetic valve, thereby allowing for safe delivery and/or deployment of the valve. However, in the absence of such an introducer, friction between the delivery device and the vasculature may be greatly increased, potentially resulting in the longitudinal compression shown in FIG. **6**, and thus potentially resulting in premature deployment of the prosthetic heart valve from the capsule or valve cover **165**.

[0064] The coiled and/or braided section **160** of the catheter assembly **150** advantageously allows for the application of high forces to unsheath the prosthetic heart valve during delivery. In some examples, the unsheath forces can be 50 pounds or more especially if the catheter is deflected in several planes at its distal end. When pull force to unsheath the valve is applied, the braid will lengthen and lock down on the coil. This prevents ovalization of the catheter assembly even if tight bends are applied and can therefore move freely on the steering catheter.

[0065] FIG. **7A** is a perspective view of a connecting ring **705** according to one or more aspects of the disclosure. FIG. **7B** is a perspective view of a valve retainer **750** according to one or more aspects of the disclosure. FIG. **7C** is a longitudinal cross-section of the connecting ring **705** of FIG. **7A** coupled with the valve retainer **750** of FIG. **7B**.

[0066] As shown in FIG. **7A**, the connecting ring **705** can be substantially cylindrical in shape and can have a distal portion **707** and a proximal portion **709**. By virtue of the substantially cylindrical shape, the distal portion **707** can define an internal surface **711** and an external surface **713**, with the external surface **713** having an external threading configured to engage with and couple to a valve cover (e.g., valve cover **550a** or **550b**). In other examples, the external surface may be unthreaded and may couple with a valve cover directly by laser welding.

[0067] The distal portion **707** of the connecting ring **705** can have a lip **715**. The lip **715** can extend radially inward relative to the internal surface **711**. The lip **715** can have a substantially planar face **715a** that can be distal-facing and positioned at a distal end of the distal portion **707**. In this regard, the lip has an internal diameter that defines an opening **714** through which a portion of the valve retainer **750** may be received.

[0068] The lip **715** can define one or more cutouts **717**. In this regard, the cutouts **717** can be defined as cutout portions of the lip face **715a** resulting in the opening **714** having an increased diameter at the cutouts as compared to at the remainder of the lip face **715a**. The cutouts **717** can have any shape and the edges of the lip face **715a** that define cutouts **717** can be any combination of linear, curved, or curvilinear. In the example of FIG. **7A**, a pair of cutouts **717** are depicted as being approximately 180 degrees from one another, while in other examples any number of cutouts **717** in any angular orientations can be implemented. As will be explained in greater detail below, the shape of the cutouts **717**, and the number and position of cutouts **717** are selected to allow for the connecting ring **705** to reversibly or selectively lock with valve retainer **750**.

[0069] The proximal portion **709** can also define an internal surface **719** and an external surface **721**. In one example, the internal surface **719** is unthreaded and the external surface **721** is unthreaded. As shown in FIG. **7C**, the internal surface **719** can define a single internal diameter and the external surface **721** can define a single outer diameter. In one example, the internal diameter of the internal surface **719** is greater than one or all of the internal diameters defined by internal surface **711**, lip **715**, or external threading **713** of the distal portion **707**.

[0070] The proximal portion **709** can define a lip **723**. The lip **723** can have a substantially planar face **723a** (labeled in FIG. **7C**) that can be proximal-facing and positioned at a distal end of the

proximal portion **709**. A channel **711a** is defined between lip **723** and lip **715**.

[0071] The valve retainer **750** can have a distal portion **752** and a proximal portion **760**. The distal portion **752** have an internal surface **754** (shown in FIG. 7C) defining at least one internal diameter. In one example, the internal surface defines an opening **756** (also referred to as a “valve seat”) configured to receive the tips of proximal petals of an atrial disk of a prosthetic valve in a collapsed state. The opening **756** can be generally cone-shaped or frustoconical, having an internal diameter that increases in the distal direction to correspond to a shape of compressed petals of the atrial disk of the prosthetic valve. In one example, the opening **756** can be the shape of a truncated cone, with the truncation being at a proximal end of the distal portion **752**. The valve retainer **750** may have a general function that is similar or identical to that of valve retainers **590a**, **590b**, which is to receive an end (e.g., an atrial or inflow end) of a prosthetic heart valve within the opening **756**, such that the end of the prosthetic heart valve is maintained in a collapsed condition by the internal surface **754**, while the remainder of the prosthetic valve is maintained in the collapsed condition by the valve cover (e.g., valve cover **550a** or **550b**).

[0072] The distal portion **752** can define one or more tabs **758** that extend radially outward from an outer surface **759** of the distal portion **752**. The tabs **758** can have any shape that is complementary to the shape of cutouts **717** and the edges that define tabs **758** can be any combination of linear, curved, or curvilinear. In the example of FIG. 7B, a pair of tabs **758** are depicted as being approximately 180 degrees from one another, while in other examples any number of tabs **758** in any angular orientations can be implemented in a fashion that is complementary to the number and orientation of cutouts **717**. As will be explained in greater detail below, the number and position of tabs **758** are selected to allow for the connecting ring **705** to reversibly or selectively lock with valve retainer **750**.

[0073] The proximal portion **760** can define an internal surface **762** and an external surface **764**. At least a portion of the external surface **764** can be a threaded region **766**.

[0074] As shown in FIG. 7C, the connecting ring **705** can engage and lock with the valve retainer **750**. To do so, the distal portion **707** of connecting ring **705** is positioned to face the proximal portion **760** of valve retainer **750**. The tabs **758** of the valve retainer **750** are first aligned with the cutouts **717** of the connecting ring **705** and then either or both of the valve retainer **750** and/or connecting ring **705** are advanced such that the tabs **758** pass through cutouts **717** and into channel **711a**. In this position, the valve retainer **750** and/or connecting ring **705** are in the unlocked position, as the tabs **758** are free to move axially in a distal direction relative to the cutouts **717**. Before rotation of either or both of the valve retainer **750** and/or connecting ring **705** from the unlocked position to a locked position, the tabs **758** are free to move axially in a distal direction relative to the cutouts **717** and thus the valve retainer **750** is free to move axially in a distal direction relative to connecting ring **705**. The tabs **758** can also move slightly proximally until the tabs **758** confront lip **723**.

[0075] After the tabs **758** pass through cutouts **717** and into channel **717a**, either or both of the valve retainer **750** and/or connecting ring **705** are then rotated to move tabs **758** along or within the channel **711a**, along internal surface **711**. This brings tabs **758** out of alignment with the cutouts **717** and brings the valve retainer **750** and the connecting ring **705** into the locked position. In one example, the rotation can be approximately 90 degrees such that the tabs **758** are 90 degrees out of alignment with the cutouts **717**. In the locked position, the lip **715** prevents removal of the tabs **758** from the channel **711a** in the distal direction (while contact between the tabs **758** and the lip **723** prevents movement of the tabs **758** in the proximal direction relative to the valve retainer **750**) without an appropriate counter rotation to once again align tabs **758** and cutouts **717**. In this locked position, the valve retainer **750** and/or connecting ring **705** are not free to (i.e., cannot) move axially relative to one another and, in particular, the valve retainer **750** cannot move axially in a distal direction relative to the connecting ring **705**.

[0076] As shown in FIG. 7C, the connecting ring **705** can be coupled or engaged with a first

catheter **790** (e.g., via adhesives, welding (e.g., laser welded), or the like) and the valve retainer **750** can be coupled or engaged with a second catheter **795**. In one example, the valve retainer **750** is screwed into the second catheter (e.g., steering catheter) and secured against rotation using adhesive. In other examples, the valve retainer **750** can be laser welded directly to the second catheter (e.g., steering catheter). The second catheter **795** can be concentrically nested within the first catheter **790**. As shown, the second catheter **795** can have an internal threading **795a** that can engage with external threading **766** of valve retainer **750**. This can occur while the connecting ring **705** is in either the unlocked or locked position. Although not depicted, the external threading of external surface **713** can be engaged with a valve cover (such as valve cover **550a** or valve cover **550b**).

[0077] In one example, the first catheter **790** can be an outer sheath (such as outer sheath **405a**) and the second catheter **795** can be a steering catheter (e.g., steering catheter **410a**) or extension catheter (e.g., extension catheter **415a**). In another example, the first catheter **790** can be an outer catheter (such as outer catheter **405b**) and the second catheter **795** can be a steering catheter (e.g., steering catheter **410b**) or intermediate catheter (e.g., intermediate catheter **415b**).

[0078] In the unlocked state, the first catheter **790** is free to move axially in a distal direction relative to the second catheter **795** since the tabs **758** are in alignment with the cutouts **717**. In the locked state, the first catheter **790** and the second catheter **795** are not free to (i.e., cannot) move axially relative to one another and, in particular, the outer catheter **790** cannot move axially in a distal direction relative to the inner catheter **795**.

[0079] In the locked state, the valve cover, which is engaged with the external threading **713** of valve retainer **750**, is axially locked to the first catheter **790**. In other examples, the valve cover may be welded directly to the valve retainer **750**. In this regard, the first catheter **790** can accommodate longitudinal compression forces exerted on the catheter assembly during delivery through the vasculature and prevent premature expression of the prosthetic heart valve. In other words, even if longitudinal compression forces are placed on a coiled or braided section of the first catheter **790**, the first catheter **790** is axially locked to the second catheter **795** (because the connecting ring **705**, which is fixed to the first catheter **790**, is axially locked to the valve retainer **750**, which in turn is fixed to the second catheter **795**). Thus, because the second catheter **795** typically does not include a coiled and/or braided section similar to the first catheter **790**, the second catheter **795** is not particularly susceptible to longitudinal compression. And since the second catheter **795** has significantly more resistance to longitudinal compression, the second catheter **795** provides resistance to longitudinal compression to the first catheter **790** when the first and second catheters are axially locked together.

[0080] FIG. **8A** is a perspective view of a connecting ring **805** and FIG. **8B** is a perspective view of a valve retainer **850** according to one or more aspects of the disclosure. FIGS. **8C-D** are transverse cross-sectional views of the connecting ring **805** of FIG. **8A** coupled with the valve retainer **850** of FIG. **8B**.

[0081] The connecting ring **805** can be similar in structure and function to the connecting ring **705**, except the cutouts **817** formed in lip **815** can define a larger arc length as compared to the cutouts **717**. Further, the internal surface **811** of connecting ring **805** can have a twist stop **811b** formed at the end of channel **811a** and between lips **815** and **723**. The twist stop **811b** can be a portion of the internal surface **811** that extends radially inward to prevent further rotation or over-rotation of the valve retainer **850** (and particularly of the tabs **858** of the valve retainer) relative to the connecting ring **805**. The tabs **858** of valve retainer **850** can define a larger arc length relative to tabs **758** and the tabs **758** can extend further radially outward than the tabs **858**. It should be understood that, in FIGS. **8A-B**, components labeled in the 700 series of numerals are intended to be similar or identical to the similarly numbered parts shown and described in connection with FIGS. **7A-B**.

[0082] As shown in FIG. **8C**, the connecting ring **805** and valve retainer **850** are in an unlocked state. In this regard, the tabs **858** of the valve retainer **850** are received within the cutouts **817** but

not yet rotated. There may be an angular tolerance AT on each side of the tabs **858** relative to the cutouts, allowing for the tabs **858** to be reliably received by the cutouts. The angular tolerance AT may be any suitable angle, and in one particular example the angle may be an approximately ± 15 to 20 degree (including approximately 18 degrees) tolerance on each side of the tabs **858** relative to the cutouts **817**. As shown in FIG. **8D**, the connecting ring **805** and valve retainer **850** are in a locked state, in which the tabs **858** have been rotated (R) approximately 115 degrees within the channel **811a** of connecting ring **805**. After the 115 degrees of rotation, the tabs **858** will confront the twist stop **811b**, preventing further rotation. Meanwhile, the tabs **858** are unaligned with the cutouts **817**, preventing the disengagement of the connecting ring **805** and valve retainer **850** and preventing the axial movement of connecting ring **805** relative to valve retainer **850**.

[0083] FIG. **9A-C** are a perspective view, front view, and rear view of a connecting ring **905** according to one or more aspects of the disclosure.

[0084] The connecting ring **905** can be similar in structure and function to the connecting ring **705**, except instead of a channel formed at internal surface **911**, the connecting ring **905** can define a plurality of tabs **958** that extend radially inward relative to lip **915**. It should be understood that, in FIGS. **9A-C**, components labeled in the 700 series of numerals are intended to be similar or identical to the similarly numbered parts shown and described in connection with FIG. **7A**.

[0085] FIGS. **9D-F** are a perspective view, front view, and rear view of a valve retainer **950** according to one or more aspects of the disclosure. It should be understood that, in FIGS. **9D-F**, components labeled in the 700 series of numerals are intended to be similar or identical to the similarly numbered parts shown and described in connection with FIG. **7B**.

[0086] Corresponding to the tabs **958**, the valve retainer **950** can define one or more channels **917a** on an exterior surface **917**. The channel **917a** can define a plurality of cutouts **917b** that can receive the tabs **958**. The connecting ring **905** and the valve retainer **950** can be aligned such that the tabs **958** are aligned with the cutouts **917b**. Once the tabs **958** are received within the cutouts **917b**, the connecting ring **905** and/or the valve retainer **950** can be rotated such that the tabs **958** rotate within the channel **917a** and are unaligned with the cutouts **917b**, thereby preventing disengagement of the connecting ring **905** and valve retainer **950**. It should be understood that, while connecting rings **705** and **805** effectively include recesses or female features (e.g., cutouts **717**, **817**) that receive the protrusions or male features (e.g., tabs **758**, **858**), the general configuration of connecting ring **905** and valve retainer **950** is reversed in the sense that the connecting ring **905** includes the protrusions or male features while the valve retainer **950** includes the recesses or female features.

[0087] FIGS. **9G-I** are a perspective view, transverse cross-sectional view, and longitudinal cross-sectional view, respectively, of the connecting ring **905** of FIGS. **9A-C** coupled with the valve retainer **950** of FIGS. **9D-F** according to one or more aspects of the disclosure. As shown, the tabs **958** are received within the channel **917a** and the connecting ring **905** and valve retainer **950** are axially locked together, thereby preventing axial movement in a distal direction of the valve retainer **950** relative to the connecting ring.

[0088] FIG. **10A** is a perspective view of a locking assembly or cone assembly **1020** of a handle assembly **1000** operable with the delivery system of FIG. **1** (e.g., in place of locking assembly or cone assembly **120**). The cone assembly **1020** can be rotated about its longitudinal axis in order to axially lock or unlock a first catheter (e.g., first catheter **790** which may be outer sheath **405a** or **405b**) relative to a second catheter (e.g., second catheter **795** which may be a steering catheter like steering catheter **410a**, **410b**, an extension catheter like extension catheter **415a**, or an intermediate catheter like intermediate catheter **415b**).

[0089] The cone assembly **1020** can include a plurality of positioning holes **1025**, and a groove **1030**. One of the positioning holes **1025** can correspond to a locked state and the other of the positioning holes **1025** can correspond to an unlocked state. In this regard, the cone assembly **1020** can be rotated relative to the housing of the handle assembly. In one example, the rotation can be approximately 90 degrees. This rotation can be guided by groove **1030**, which can receive a portion

of the housing of the handle assembly, ensuring that cone assembly **1020** is axially fixed relative to the handle assembly.

[0090] FIGS. **10B-C** are side views of the cone assembly **1020** of FIG. **10A** in the unlocked and locked state, respectively. As shown in FIGS. **10A-B**, a screw **1035** of the handle assembly is received within the positioning hole **1025**, thereby securing the cone assembly **1020** in the unlocked state. To move to the locked state, the screw **1035** is rotated to disengage from the positioning hole **1025** and the cone assembly **1020** is rotated. In other examples, the screw can be replaced with a spring-loaded pin configured to be snap into the positions holes **1025** when released and to be pulled outwardly to disengage. The cone assembly **1020** and/or handle assembly can optionally include stopping elements to prevent over rotation. In one example, the cone assembly **1020** is rotated until the locked state is reached. Upon reaching the locked state, the screw **1035** can be rotated to engage with the next positioning hole **1025**, thereby securing the cone assembly in the locked state relative to the handle assembly. As shown in FIG. **1**, the cone assembly **1020** can be operatively coupled to the catheter assembly such that rotation of the cone assembly **1020** can cause a corresponding rotation of the outer catheter (e.g., first catheter **790** or outer sheath **405a** or **405b**), allowing for the cone assembly **1020** to actuate between the locked and unlocked of the catheter assembly. In this regard, rotation of the cone assembly **1020** and outer catheter can cause relative rotation of the tabs and/or channels of the connecting ring and/or valve retainer, thereby providing the necessary axial alignment and/or unalignment.

[0091] FIG. **11** is a schematic cutaway representation of a patient's heart H and a delivery route that may be followed by catheter assembly **1100** to reach the native mitral valve annulus MVA. It should be understood that catheter assembly **1100** may be representative of any of the catheter assemblies previously described herein. Using a transfemoral approach, catheter assembly **1100** may be inserted over guidewire **1110** into the patient's femoral vein and advanced through the inferior vena cava IVC to the right atrium RA. In a subsequent transseptal procedure, catheter assembly **1100** is advanced through a puncture in intra-atrial septum S into the left atrium LA. Once the catheter assembly **1100** is positioned at the desired location relative to the mitral valve annulus MVA, the catheter may be operated (e.g., by rotating valve cover retraction knob **115** to withdraw the valve cover **160** relative to the prosthetic heart valve maintained in a collapsed condition within the valve cover) to deploy a prosthetic mitral valve into the mitral valve annulus MVA.

[0092] In other implementations, such as for procedures associated with a tricuspid valve, catheter assembly **1100** may be advanced through the inferior vena cava IVC and into the right atrium RA, where it may then be positioned and used to perform the procedure related to the tricuspid valve. While many of the examples described herein relate to delivery to the native mitral valve annulus, one or more embodiments may be utilized in other cardiac procedures, including those involving the tricuspid valve or other cardiac valves.

[0093] Although one preferred method for accessing a targeted cardiac valve annulus is a transfemoral approach, it will be understood that the embodiments described herein may also be utilized where alternative approaches are used. For example, embodiments described herein may be utilized in a transjugular approach, transapical approach, transradial approach or other suitable approaches to the targeted anatomy. For procedures related to the mitral valve or tricuspid valve, the delivery of the prosthetic heart valve or other medical device is preferably carried out from an atrial aspect (i.e., with the distal end of catheter assembly **1100** positioned within the atrium superior to the targeted valve). The illustrated embodiments are shown from such an atrial aspect. However, it will be understood that the delivery of the medical devices described herein may also be carried out from a ventricular aspect.

[0094] Additional details regarding delivery systems and devices that may be utilized in conjunction with the components and features described herein are described in U.S. Patent Publication Nos. 2018/0028177A1, 2018/0092744A1, and 2020/0155804, the disclosures of which

are hereby incorporated by reference herein.

[0095] FIG. 12 is a flow chart depicting a method of operating a handle assembly during delivery of the prosthetic heart valve to the mitral valve annulus.

[0096] At block 1210, the delivery catheter is advanced through the vasculature. For example, the catheter assembly may be inserted into the patient's femoral vein and advanced until it reaches an opening of the inferior vena cava leading to the right atrium. The forces can be relatively high during insertion through the femoral vein until the valve cover reaches the inferior vena cava, where the larger diameter of the inferior vena cava reduces or eliminates frictional forces on the valve cover. While this advancing is occurring, a prosthetic heart valve (e.g., a self-expanding prosthetic mitral valve) may be maintained in a collapsed condition within a valve cover of the catheter assembly. Importantly, in some embodiments, an introducer sheath or introducer device may be omitted from the procedure such that the catheter assembly is inserted directly into the patient's femoral vein without a separate introducer catheter within the femoral vein through which the catheter assembly is advanced. During block 1210, the catheter assembly can be in the locked state and thus the valve cover and outer sheath (e.g., first catheter 790) can be axially locked relative to an inner catheter (e.g., second catheter 795, which may be one of the steering catheters described above, or the extension or intermediate catheter described above). In the locked state, force can be applied to the catheter assembly to push the assembly through the vasculature without the risk of premature unsheathing and/or expression of the prosthetic heart valve. In this locked state, the second catheter can be an extension or steering catheter, which can accommodate the compression forces exerted during delivery, and thus provide stability to the first catheter (or outer sheath) to prevent collapsing of any coiled or braided sections of the first catheter.

[0097] At block 1220, second catheter can be unlocked relative to the first catheter (e.g. by unlocking the connection ring relative to the valve retainer). This can be done by rotating the cone assembly, which can disengage the respective tabs and/or channels of the connector ring and/or valve retainer, thereby allowing for axial movement of the valve retainer relative to the connecting ring. The unlocking can occur at any time during advancement of the catheter assembly, and in one example occurs at or near the opening of the inferior vena cava leading to the right atrium. In another example, the unlocking can occur when the valve cover is in the vena cava and is in a relatively straight (unbent or undeflected) position, as the rotational forces for unlocking are minimized in the straight position as compared to a bent or deflected position. It may be desirable to keep the catheter assembly in the locked condition while the catheter assembly is in the femoral vein to avoid the premature valve deployment described above. However, while the catheter assembly is in the locked condition and the first catheter is locked to the second catheter, it may be difficult to bend the catheter to any significant degree which may be required to guide the catheter through the heart. In many patient anatomies, the femoral vein has a relatively non-tortuous path between a typical entry point in the groin to the point at which the inferior vena cava IVC opens into the right atrium RA. Thus, the more difficult bending operation of the catheter assembly may not be necessary prior to the catheter assembly exiting the inferior vena cava IVC. But upon exiting the inferior vena cava IVC, the bending or steering operation may be highly desirable and thus unlocking the catheter assembly at or near the inferior vena cava IVC is one suitable option.

[0098] At block 1230, with the valve retainer unlocked relative to the connecting ring, the prosthetic heart valve can be delivered to the native mitral valve annulus, for example through the right atrium RA, septum S and left atrium LA. Various steering or bending operations may be performed to guide the steering catheter along this path and to eventually position the valve cover near or in the mitral valve annulus MVA. When the valve cover is in the final desired position, the prosthetic heart valve may be deployed by retracting the valve cover relative to the catheter to which the valve retainer is connected. Since the valve retainer is in the unlocked state at this point, the valve cover is free to retract axially relative to the valve retainer, and thus the prosthetic heart valve is free to express from the valve cover of the catheter assembly and into the native mitral

valve annulus.

[0099] At block **1240**, the catheter assembly is retracted through the vasculature and removed from the patient. During removal, the catheter assembly can be in the locked or unlocked state, although it may be preferable to keep the catheter assembly in the unlocked state for easier bending if needed.

[0100] Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A delivery system for delivering a prosthetic heart valve, the delivery system comprising: a handle assembly; a catheter assembly extending from the handle assembly, the catheter assembly having a first catheter and a second catheter positioned radially inside the first catheter; a connecting ring fixed to a distal end portion of the first catheter; and a valve retainer fixed to a distal end portion of the second catheter, the valve retainer configured to receive at least a portion of the prosthetic heart valve therein, wherein the catheter assembly has i) a locked state in which the valve retainer is axially locked relative to the connecting ring, and ii) an unlocked state in which the valve retainer is axially movable relative to the connecting ring, the handle assembly being configured to actuate the catheter assembly between the locked state and the unlocked state.
2. The delivery system of claim 1, wherein the first catheter has a coiled or braided section positioned proximal to the connecting ring.
3. The delivery system of claim 2, wherein the second catheter has a greater resistance to longitudinal compression compared to the first catheter.
4. The delivery system of claim 3, wherein: the first catheter is an outer sheath of the catheter assembly; and the second catheter is a steering catheter or an extension catheter.
5. The delivery system of claim 3, wherein: the first catheter is an outer sheath of the catheter assembly; and the second catheter a steering catheter or an intermediate catheter.
6. The delivery system of claim 3, wherein the connecting ring defines a slot configured to receive a tab of the valve retainer.
7. The delivery system of claim 3, wherein the valve retainer defines a slot configured to receive a tab of the connecting ring.
8. The delivery system of claim 3, wherein the handle assembly includes a locking assembly such that rotation of the locking assembly relative to other components of the handle assembly actuates the catheter assembly between the locked state and the unlocked state.
9. The delivery system of claim 1, wherein the catheter assembly includes a valve cover fixed to the connecting ring, the valve cover configured to maintain at least part of the prosthetic heart valve in a collapsed condition.
10. The delivery system of claim 9, wherein in the locked state of the catheter assembly, the valve cover is axially locked relative to the valve retainer, thereby preventing premature expression of the prosthetic heart valve from the valve cover.
11. The delivery system of claim 10, wherein in the unlocked state of the catheter assembly, the valve cover is configured to retract proximally along with the first catheter relative to the second catheter and relative to the valve retainer, thereby allowing the for expression of the prosthetic heart valve from the valve cover.
12. The delivery system of claim 9, wherein the valve cover is threadedly coupled to the connecting ring.
13. The delivery system of claim 9, wherein the wherein the handle assembly includes (i) a valve

cover retraction knob and (ii) a locking assembly, such that rotation of the valve cover retraction knob in the unlocked state of the catheter assembly proximally withdraws the valve cover relative to the valve retainer, and rotation of the locking assembly relative to other components of the handle assembly actuates the catheter assembly between the locked state and the unlocked state, the valve cover retraction knob being positioned distal to the locking assembly.

14. The delivery system of claim 1, wherein the valve retainer is threadedly coupled to the second catheter.

15. A method of delivering a prosthetic heart valve, comprising: advancing a catheter assembly through a vasculature while the catheter assembly is in a locked state in which a first catheter is axially locked relative to a second catheter nested within the first catheter, and while the prosthetic heart valve is maintained in a collapsed condition within a valve cover coupled to a distal end of the first catheter; actuating the catheter assembly to an unlocked state, thereby allowing for axial movement of the first catheter relative to the second catheter; and retracting the valve cover and the first catheter relative to the second catheter to deploy the prosthetic heart valve into a native mitral valve annulus.

16. The method of claim 15, wherein actuating the catheter assembly to the unlocked state is performed when a distal end of the catheter assembly is positioned within or adjacent to an opening of an inferior vena cava leading to a right atrium.

17. The method of claim 15, wherein actuating the catheter assembly to the unlocked state includes rotating a locking assembly of a handle assembly, the catheter assembly extending from the handle assembly.

18. The method of claim 15, wherein the catheter assembly includes a connecting ring that couples the valve cover to the distal end of the first catheter, and a valve retainer coupled to a distal end of the second catheter.

19. The method of claim 18, wherein actuating the catheter assembly to the unlocked state includes rotating the first catheter relative to the second catheter such that the valve retainer rotates relative to the connecting ring.
